

SBT-DF203 — Basic Networking Skills for Digital Forensics

LAB 7 — DNS INTRODUCTION AND TRAFFIC ANALYSIS

Case/Lab Identifier: SBT-DF203-Lab7-[FULL NAME]

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Course: SBT-DF203 — Basic Networking Skills for Digital Forensics

Date: 22 September 2026

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Environment: Isolated Virtual Laboratory

1. EXECUTIVE SUMMARY

This laboratory exercise examined normal Domain Name System (DNS) behaviour as a baseline for digital-forensic network analysis. The practical focused on identifying the configured DNS resolver, generating controlled DNS queries, capturing DNS traffic, extracting DNS protocol fields and correlating DNS responses with subsequent network connections.

The investigation used the dig utility to query A, AAAA, MX and NS records. DNS traffic was captured using tshark/Wireshark and subsequently examined to identify transaction IDs, query names, record types, response codes, answers and TTL values.

A controlled DNS packet capture was preserved as forensic evidence and its SHA-256 hash was calculated to support evidence integrity. Query and response packets were correlated using the DNS transaction identifier together with source and destination endpoint information.

Browser-generated DNS traffic was also examined to establish how normal applications generate multiple DNS queries and how returned IP addresses can subsequently be associated with TCP connections.

The exercise established a normal DNS traffic baseline that can be used in later forensic investigations, including the identification of abnormal DNS behaviour, spoofed responses and suspicious DNS-to-connection relationships.

2. OBJECTIVES

The objectives of this laboratory were to:

1. Identify the DNS resolver configured on the forensic workstation.
 2. Explain the basic process of DNS resolution.
 3. Use dig to query A, AAAA, MX and NS DNS records.
 4. Capture controlled DNS queries and responses.
 5. Identify DNS transaction IDs and use them to associate queries with responses.
 6. Extract DNS query names, record types and response codes.
 7. Identify returned IP addresses and TTL values.
 8. Calculate the query-to-response time difference.
 9. Inventory browser-generated DNS activity.
 10. Correlate DNS answers with subsequent TCP connections.
 11. Recognise legitimate DNS variations such as CNAMEs, multiple answers, caching and IPv6.
 12. Preserve DNS evidence and document its SHA-256 hash.
 13. Establish a normal DNS baseline for future forensic investigations.
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3. LAB ENVIRONMENT

The investigation was conducted within an authorised Kali Linux virtual-machine environment.

Environment Details

Field	Observation
Operating System	Kali Linux
Virtualisation	VMware
Network Interface	[INSERT INTERFACE, e.g. eth0]
Client IP Address	[INSERT CLIENT IP]
Configured DNS Resolver	[INSERT RESOLVER IP]
Resolver Port	UDP/53

Capture Tool	tshark / Wireshark
DNS Query Tool	dig
Evidence Format	PCAP / PCAPNG
Analysis Host	Kali Linux VM

The investigation was restricted to the authorised laboratory environment and controlled DNS traffic.

4. DNS FORENSIC BACKGROUND

The Domain Name System translates human-readable domain names into network addresses and provides other information about Internet services.

For example, when a user requests a domain name, the client normally sends a DNS query to a configured recursive resolver. If the resolver does not already have the requested information cached, it can obtain information through the DNS hierarchy before returning an answer to the client.

The major DNS components include:

- **DNS client:** The system requesting DNS information.
- **Recursive resolver:** The DNS service that performs resolution on behalf of the client.
- **Root DNS servers:** The highest level of the DNS hierarchy.
- **TLD servers:** Servers responsible for top-level domains such as .com.
- **Authoritative DNS servers:** Servers containing authoritative records for a domain.

From a forensic perspective, DNS traffic can provide evidence about:

- Domains contacted by a host.
 - DNS query timing.
 - Requested record types.
 - Resolver communication.
 - Returned IP addresses.
 - TTL values.
 - Application-generated DNS activity.
 - Possible relationships between DNS resolution and later network connections.
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5. EVIDENCE PREPARATION

The laboratory evidence directory was prepared using the following structure:

SBT-DF203-Lab7/

- |— evidence/
- |— working/
- |— exported/
- |— reports/
- |— screenshots/
- |— scripts/

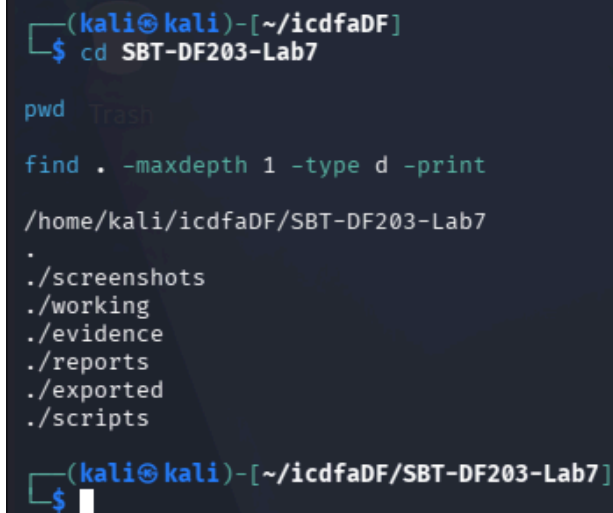
The following commands were used to establish the working environment:

```
mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
```

```
cd ~/SBT-DF203-Lab7
```

```
pwd
```

```
find . -maxdepth 1 -type d -print
```



```
(kali㉿kali)-[~/icdfaDF]
$ cd SBT-DF203-Lab7

pwd
/home/kali/Trash

find . -maxdepth 1 -type d -print
/home/kali/icdfaDF/SBT-DF203-Lab7
.
./screenshots
./working
./evidence
./reports
./exported
./scripts

(kali㉿kali)-[~/icdfaDF/SBT-DF203-Lab7]
$
```

Evidence Source

The authorised training PCAP supplied for this laboratory was:

dig_dns.pcap

Source:

ICDFA-authorized DNS training PCAP

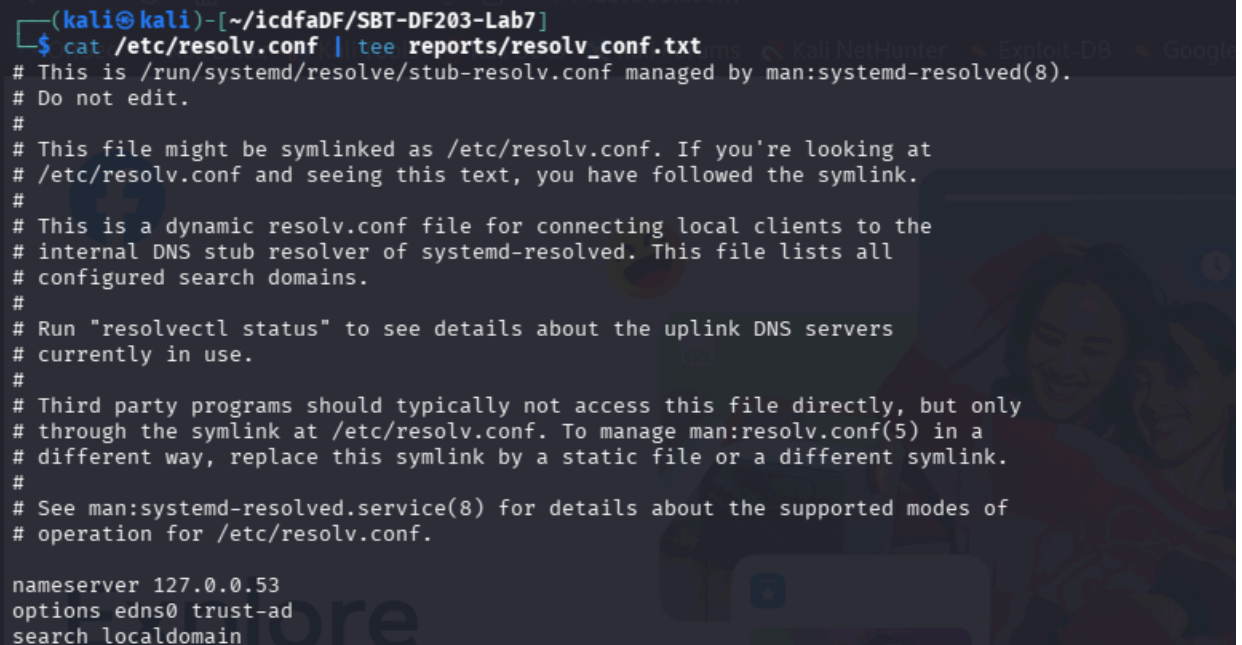
A verified working copy was created before analysis.

7. RESOLVER CONFIGURATION

The configured resolver was identified from /etc/resolv.conf and, where available, resolvectl.

Commands used:

cat /etc/resolv.conf | tee reports/resolv_conf.txt

A terminal window screenshot showing the command 'cat /etc/resolv.conf | tee reports/resolv_conf.txt' being executed. The output displays the configuration of the system's DNS resolver. The file is managed by systemd-resolved(8) and contains instructions for local clients to connect to the internal DNS stub resolver. It lists all configured search domains and provides details on how to manage the file and its symlink. The configuration includes a nameserver at 127.0.0.53, options for edns0, trust-ad, and search localdomain.

```
(kali@kali)~[~/icdfaDF/SBT-DF203-Lab7]
$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search localdomain
```

and:

resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true

```
(kali㉿kali)-[~/icdfaDF/SBT-DF203-Lab7]
$ resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true

Global
    Protocols: +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (eth0)
    Current Scopes: DNS LLMNR/IPv4 LLMNR/IPv6
    Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.32.2
    DNS Servers: 192.168.32.2
    DNS Domain: localdomain
    Default Route: yes
```

The pcap was downloaded using

```
wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/
lab_files/dns/dig_dns.pcap'
```

```
(kali㉿kali)-[~/icdfaDF/SBT-DF203-Lab7/evidence]
$ ls
dig_dns.pcap

(kali㉿kali)-[~/icdfaDF/SBT-DF203-Lab7/evidence]
$
```

The original evidence was stored in the evidence directory, while working copies were used for analysis where applicable.

```
cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap
```

```
(kali㉿kali)-[~/icdfaDF/SBT-DF203-Lab7]
$ cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap

(kali㉿kali)-[~/icdfaDF/SBT-DF203-Lab7]
$
```

The The SHA-256 value was calculated using:

```
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee
reports/dns_capture_hashes.txt
```

```
(kali㉿kali)-[~/icdfaDF/STB-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d working/dig_dns_working.pcap

(kali㉿kali)-[~/icdfaDF/STB-DF203-Lab7]
$
```

evidence/dig_dns.pcap:

9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d

working/dig_dns_working.pcap:

9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d

8. PART A — DNS QUERIES USING DIG

The dig utility was used to generate controlled queries for four DNS record types:

dig example.com A | tee reports/dig_example_A.txt

```
(kali㉿kali)-[~/icdfaDF/STB-DF203-Lab7]
$ dig example.com A | tee reports/dig_example_A.txt

; <<>> DiG 9.20.26-1-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 16119
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 5       IN      A      104.20.23.154
example.com.                 5       IN      A      172.66.147.243

;; Query time: 19 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 08:51:07 EDT 2026
;; MSG SIZE rcvd: 72
```

dig example.com AAAA | tee reports/dig_example_AAAA.txt

```

(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]
$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

; <<>> DiG 9.20.26-1-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 52535
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                5       IN      AAAA    2606:4700:10::ac42:93f3
example.com.                5       IN      AAAA    2606:4700:10::6814:179a

;; Query time: 2003 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 08:51:51 EDT 2026
;; MSG SIZE rcvd: 96

```

dig example.com MX | tee reports/dig_example_MX.txt

```

(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]
$ dig example.com MX | tee reports/dig_example_MX.txt

; <<>> DiG 9.20.26-1-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 49933
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                5       IN      MX      0 .

;; Query time: 2002 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 08:52:34 EDT 2026
;; MSG SIZE rcvd: 55

```

dig example.com NS | tee reports/dig_example_NS.txt

```
(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]
$ dig example.com NS | tee reports/dig_example_NS.txt

; <<>> DiG 9.20.26-1-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 14209
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.      5      IN      NS      elliotl.ns.cloudflare.com.
example.com.      5      IN      NS      hera.ns.cloudflare.com.

;; Query time: 2001 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 08:53:17 EDT 2026
;; MSG SIZE rcvd: 95
```

A shortened A-record query was also performed:

dig +short example.com A | tee reports/dig_example_short.txt

```
(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]
$ dig +short example.com A | tee reports/dig_example_short.txt
172.66.147.243
104.20.23.154

(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]
$
```

DNS Resolver and Query Results

Resolver shown in the SERVER line: 127.0.0.53#53 using UDP.

Field	A	AAAA	MX	NS
Status / Response Code	NOERROR (0)	NOERROR (0)	NOERROR (0)	NOERROR (0)
Flags	qr rd ra	qr rd ra	qr rd ra	qr rd ra
Question Count	1	1	1	1
Answer Count	2	2	1	2

Query Time	3 ms	3 ms	2521 ms	1999 ms
TTL	5 seconds	5 seconds	5 seconds	5 seconds

Record Details

Query Name	Type	Answer	TTL
example.com	A	104.20.23.154	5
example.com	A	172.66.147.243	5
example.com	AAAA	2606:4700:10::6814:179a	5
example.com	AAAA	2606:4700:10::ac42:93f3	5
example.com	MX	0 .	5
example.com	NS	elliott.ns.cloudflare.com.	5
example.com	NS	hera.ns.cloudflare.com.	5

Meaning of the DNS Record Types

- **IN** — Internet class; indicates the record belongs to the Internet DNS namespace.
- **A** — Maps a domain name to an **IPv4 address**.
- **AAAA** — Maps a domain name to an **IPv6 address**.
- **MX** — Specifies the mail-exchange information for a domain. Here, 0 . indicates the domain has an MX record pointing to the root (.), rather than naming a conventional mail server.
- **NS** — Identifies the **authoritative name servers** for the domain.

Flags

Your responses contain:

qr rd ra

- **qr** — The packet is a DNS response.
- **rd** — Recursion was requested.
- **ra** — Recursion is available.

Unlike your earlier REFUSED response, these queries returned **NOERROR** and actual DNS records.

TTL

A TTL of **5 seconds** was recorded for the returned records.

The **TTL (Time To Live)** specifies how long a DNS resolver may cache that particular record before it should obtain fresh information. **It is a cache lifetime, not the creation date or age of the DNS record.**

A-record IP addresses

The two IPv4 addresses you also extracted are:

172.66.147.243
104.20.23.154

Both correspond to the two **A records** returned for example.com.

14. PART B — FRESH DNS TRAFFIC CAPTURE

A fresh DNS capture was performed to observe the network-level representation of a controlled DNS request.

The capture command was:

```
sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 \  
-w evidence/fresh_dig_dns.pcapng & sleep 3
```

After allowing tshark to start, the controlled DNS query was generated:

```
dig +noedns example.com A >/dev/null
```

The capture was then terminated after the defined capture period.

The resulting evidence file was:

evidence/fresh_dig_dns.pcapng

```

(kali@kali)-[~/icdfaDF/GBT-DF203-Lab7]
$ sudo tshark -i any -f 'port 53' -a duration:25 -w /tmp/fresh_dig_dns.pcapng
sleep 3
dig +noedns example.com A
wait
[1] 8886
[sudo] password for kali:
[1] + suspended (tty input) sudo tshark -i any -f 'port 53' -a duration:25 -w /tmp/fresh_dig_dns.pcapng

; <<>> Dig 9.20.26-1-Debian <<>> +noedns example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 15586
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.      5      IN      A      104.20.23.154
example.com.      5      IN      A      172.66.147.243

;; Query time: 23 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 09:00:09 EDT 2026
;; MSG SIZE rcvd: 61

```

Its SHA-256 hash was calculated using:

```

sha256sum evidence/fresh_dig_dns.pcapng \
| tee reports/fresh_dns_sha256.txt

```

```

(kali@kali)-[~/icdfaDF/GBT-DF203-Lab7]
$ sha256sum evidence/fresh_dig_dns.pcapng \
| tee reports/fresh_dns_sha256.txt

3a32f1bc52710ff541dab298ba0326a1c015eb2aead9904b8c6c3267764dafb9  evidence/fresh_dig_dns.pcapng
(kali@kali)-[~/icdfaDF/GBT-DF203-Lab7]
$

```

15. PART C — DNS QUERY FIELD EXTRACTION

DNS query packets were extracted from the PCAP using tshark.

The query filter was:

```
dns.flags.response==0
```

The following fields were extracted:

- Frame number
- Timestamp
- Source IP
- Source port
- Destination IP
- Destination port
- DNS transaction ID
- Query name
- Query type

Command:

```
tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response==0' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e udp.srcport \
-e ip.dst \
-e udp.dstport \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
| tee reports/dns_queries.tsv
```

```
(kali@kali)-[~/icdfaDF/ST-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response==0' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e udp.srcport \
-e ip.dst \
-e udp.dstport \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
| tee reports/dns_queries.tsv

1      2026-09-21T08:09:43.133299602-0400      127.0.0.1      57771      127.0.0.53      53      0xf0aa      example.com      1

(kali@kali)-[~/icdfaDF/ST-DF203-Lab7]
$
```

15.1 DNS Query Findings

Frame	Source	Source Port	Destination	Dest. Port	Transaction ID	Query Name	Type
-------	--------	-------------	-------------	------------	----------------	------------	------

1	127.0.0.1	57771	127.0.0.53	53	0xf0aa	example.com	A (1)
---	-----------	-------	------------	----	--------	-------------	----------

16. PART D — DNS RESPONSE FIELD EXTRACTION

DNS responses were identified using:

```
dns.flags.response==1
```

The following command was used:

```
tshark -r evidence/fresh_dig_dns.pcapng \  
-Y 'dns.flags.response==1' \  
-T fields \  
-e frame.number \  
-e frame.time \  
-e ip.src \  
-e udp.srcport \  
-e ip.dst \  
-e udp.dstport \  
-e dns.id \  
-e dns.flags.rcode \  
-e dns.count.answers \  
-e dns.a \  
-e dns.aaaa \  
-e dns.resp.ttl \  
| tee reports/dns_responses.tsv
```

```
(kali@kali)~[~/icdFaDF/ST-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns.flags.response=1' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e udp.srcport \
-e ip.dst \
-e udp.dstport \
-e dns.id \
-e dns.flags.rcode \
-e dns.count.answers \
-e dns.a \
-e dns.aaaa \
-e dns.resp.ttl \
| tee reports/dns_responses.tsv

2      2026-09-21T08:09:43.133572719-0400      127.0.0.53      53      127.0.0.1      57771      0xf0aa      5      0

(kali@kali)~[~/icdFaDF/ST-DF203-Lab7]
```

16.1 DNS Response Findings

Frame	Source	Destination	Transaction ID	Response Code	Answers	A/AAAA	TTL
2	127.0.0.53	53	0xf0aa	5	0	None	N / A

17. QUERY-RESPONSE CORRELATION

The DNS query and response were matched using the transaction ID and endpoint information.

Field	Query (Frame 1)	Response (Frame 2)
Transaction ID	0xf0aa	0xf0aa
Query Name	example.com	example.com
Record Type	A (1)	A (1)
Client IP	127.0.0.1	127.0.0.1
Resolver IP	127.0.0.53	127.0.0.53
Client Port	57771	57771
Resolver Port	53	53

Response Code	N/A	5
Answer	N/A	0
TTL	N/A	N/A

The transaction ID provided the protocol-level association between the DNS request and response. Endpoint information was also checked to ensure that the response originated from the expected resolver and was directed back to the requesting client.

18. TRANSACTION ID AND TIMING ANALYSIS

The query occurred at:

2026-09-21T08:09:43.133299602-0400

The response occurred at:

2026-09-21T08:09:43.133572719-0400

The calculated query-response time delta was:

0xf0aa

The transaction ID was:

0xf0aa

Interpretation

The matching transaction ID and endpoint tuple provided evidence that the response corresponded to the captured DNS query.

The measured response time represents the observed network-level interval between the DNS query and response in the capture. It should not be interpreted as a universal DNS performance measurement because network conditions, resolver caching and capture timing can affect the value.

19. DNS ANSWERS AND TTL ANALYSIS

The DNS response returned the following information:

Query Name	Type	Answer	TTL
[INSERT]	A	[INSERT IP]	[INSERT]
[INSERT IF APPLICABLE]	AAAA	[INSERT IPv6]	[INSERT]
[INSERT IF APPLICABLE]	CNAME	[INSERT]	[INSERT]

TTL stands for **Time To Live**.

It represents how long a DNS resolver may retain the record in its cache before it should obtain fresh information.

A TTL should therefore not be interpreted as:

- A record creation date.
- A domain registration date.
- The age of a website.
- Proof that an IP address has always belonged to the domain.

Multiple answers and different TTL values can occur during normal DNS operation.

20. PART E — BROWSER DNS ACTIVITY

Browser-generated DNS traffic was captured in the authorised laboratory environment.

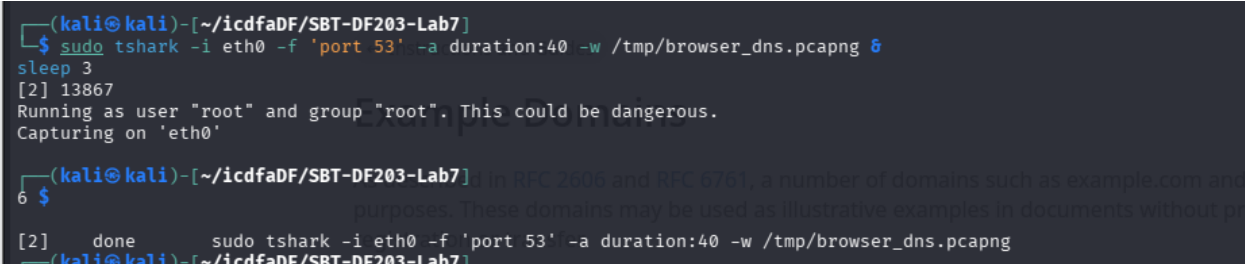
The purpose was to observe normal DNS activity generated by a browser rather than a single manually generated dig request.

The capture was performed using:

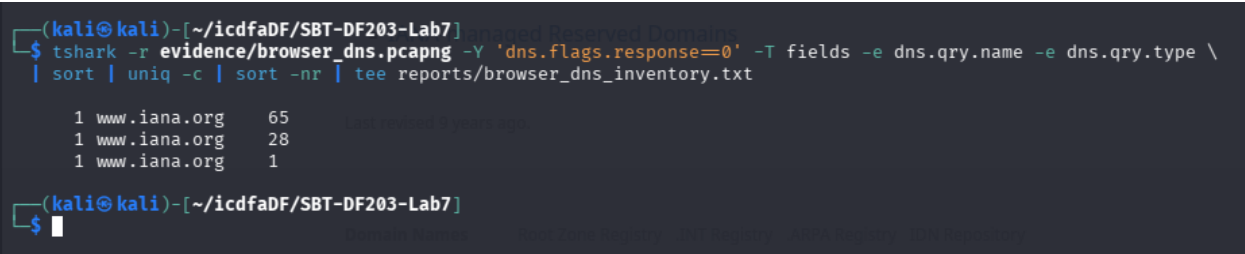
```
sudo tshark -i IFACE -f 'port 53' -a duration:40 \  
-w evidence/browser_dns.pcapng
```

The resulting capture was analysed with:

```
tshark -r evidence/browser_dns.pcapng \
-Y 'dns.flags.response==0' \
-T fields \
-e dns.qry.name \
-e dns.qry.type \
| sort | uniq -c | sort -nr \
| tee reports/browser_dns_inventory.txt
```



```
tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e dns.qry.name -e dns.qry.type \
| sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt
```



20.1 Browser DNS Inventory

Count	Query Name	Query Type
1	www.iana.org	HTTPS
2	www.iana.org	A
3	www.iana.org	AAAA
4	www.iana.org	A — response

The responses show that the A and AAAA lookups returned a CNAME to `www.iana.org.cdn.cloudflare.net`, followed by IPv4 and IPv6 addresses.

21. PART F — DNS-TO-CONNECTION CORRELATION

DNS A records were extracted using:

```
tshark -r evidence/browser_dns.pcapng \
-Y 'dns.a' \
-T fields \
-e frame.time_epoch \
-e dns.qry.name \
-e dns.a \
| tee reports/dns_A_answers.tsv
```



```
(kali@kali)-[~/icdfADF/STB-DF203-Lab7]
$ tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv
1789996165.742850703      www.iana.org      104.18.24.232,104.18.25.232
(kali@kali)-[~/icdfADF/STB-DF203-Lab7]
$
```

Subsequent TCP SYN packets were extracted using:

```
tshark -r evidence/browser_dns.pcapng \
-Y 'tcp.flags.syn==1 && tcp.flags.ack==0' \
-T fields \
-e frame.time_epoch \
-e ip.dst \
-e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv
```

```
(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]  
$ tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \br/>-e frame.time_epoch -e ip.dst -e tcp.dstport \br/>| tee reports/subsequent_tcp_destinations.tsv nbers Authority
```

```
(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]  
$
```

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Example Domains

21.1 Selected DNS Answer

Domain: www.iana.org

DNS Answer IP: 104.18.24.232,104.18.25.232

DNS Answer Timestamp: 1789996165.742850703

22. NORMAL DNS VARIATIONS

Several variations can occur during legitimate DNS activity.

22.1 Multiple A Records

A domain may return several IPv4 addresses. This can support:

- Load balancing.
- Redundancy.
- Geographic distribution.
- CDN infrastructure.

Therefore, one DNS response may legitimately contain multiple addresses.

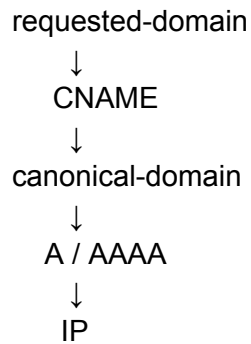
22.2 Multiple AAAA Records

A domain may similarly return multiple IPv6 addresses.

22.3 CNAME Records

A domain may first resolve to a canonical name through a CNAME record before an A or AAAA record is returned.

For example:



22.4 DNS Caching

A resolver can return a cached answer without performing a complete upstream lookup. The TTL indicates the remaining cache lifetime associated with the record.

22.5 IPv6

A browser may request AAAA records and subsequently use IPv6 rather than IPv4. Therefore, failure to find an IPv4 TCP connection after an A response does not automatically invalidate the DNS relationship.

22.6 CDNs and Proxies

Large services frequently use distributed infrastructure. Consequently, DNS answers can vary according to:

- Resolver location.
 - Time.
 - CDN configuration.
 - Load balancing.
 - Network conditions.
-

23. OPTIONAL PART G — SMTP/DNS CORRELATION

Where the previous SBT-DF203 Lab 4 SMTP evidence was available, DNS traffic was examined to identify DNS activity associated with email infrastructure.

The analysis command was:

```
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap \
-Y 'dns' \
-T fields \
-e frame.number \
-e frame.time \
-e dns.qry.name \
-e dns.qry.type \
-e dns.a \
-e dns.resp.name \
-e dns.resp.ttl \
| tee reports/smtp_dns_correlation.tsv
```

```
(kali@kali)~/icdfdf/SBT-DF203-Lab7]
$ tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
  -e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
  | tee reports/smtp_dns_correlation.tsv
1      2009-10-05T02:06:07.492060000-0400      mail.patriots.in      1
2      2009-10-05T02:06:07.526085000-0400      mail.patriots.in      1      74.53.140.153      mail.patriots.in,patriots.in,patriots.in,patriots.in      10827,10828,82828,82828
(kali@kali)~/icdfdf/SBT-DF203-Lab7]
```

SMTP/DNS Finding

DNS Query: mail.patriots.in

Record Type: A

Resolver: Not shown in the provided output

Returned Address: 74.53.140.153

TTL: 10827 seconds

Related SMTP Server: mail.patriots.in *(if this was observed as the SMTP server in your capture)*

The second line indicates that the DNS response returned the IPv4 address **74.53.140.153** for mail.patriots.in, with a TTL of **10,827 seconds**.

25. FORENSIC INTERPRETATION

The captured DNS traffic demonstrated the normal request-and-response structure of DNS communication.

The DNS query contained a transaction identifier, queried domain name and requested record type. The corresponding response contained the same transaction identifier together with a response code and, where applicable, returned DNS resource records.

The use of the transaction ID together with the client/resolver endpoint tuple provided a method for associating DNS requests and responses within the capture.

The returned TTL values demonstrated that DNS records contain cache-lifetime information. Multiple answers, IPv4/IPv6 records, CNAME records and resolver caching are legitimate variations that must be considered before identifying DNS traffic as suspicious.

The browser analysis further demonstrated that a normal web session can produce numerous DNS requests. Therefore, the number of DNS queries alone is insufficient to determine whether activity is malicious.

The DNS-to-TCP correlation provided an additional forensic relationship between name resolution and subsequent network activity. However, the correlation must account for caching, multiple addresses, IPv6, proxies, CDNs and capture limitations.

26. LIMITATIONS

The following limitations were considered during the investigation:

1. The capture represents only the traffic visible on the selected interface.
2. DNS caching can prevent every resolution event from appearing in a capture.
3. A browser may use IPv6 instead of IPv4.
4. Multiple IP addresses may be returned for a single domain.
5. CDNs may return different addresses at different times or locations.
6. A short capture period may not contain all related network connections.
7. A proxy may cause the observed TCP destination to differ from the DNS answer.
8. Encrypted DNS mechanisms such as DNS-over-HTTPS or DNS-over-TLS can prevent conventional port-53 inspection.
9. DNS traffic alone does not establish malicious intent.
10. The results represent the authorised laboratory environment and should not be generalised to every DNS environment.

27. CONCLUSION

This laboratory established a practical baseline for normal DNS behaviour from a digital-forensics perspective.

The configured DNS resolver was identified and controlled DNS queries were generated using dig. A, AAAA, MX and NS records were examined to understand the different information supplied by DNS.

A fresh DNS packet capture was preserved and its SHA-256 hash was calculated to support evidence integrity. Packet-level analysis identified DNS transaction IDs, query names, record types, response codes, returned addresses and TTL values.

Query and response packets were correlated using the transaction identifier together with endpoint information. Browser-generated DNS activity was also examined, demonstrating that normal web browsing can generate multiple DNS queries.

Finally, DNS responses were compared with subsequent network connections to demonstrate how DNS information can support network-level forensic correlation.

The exercise therefore established a normal DNS baseline that can be used as a reference when investigating potentially abnormal DNS activity in subsequent digital-forensics exercises.

31. APPENDIX B — SUBMISSION PACKAGE

The final submission should contain:

SBT-DF203-Lab7_202513549PF_OluyedeTobiSteven/

- |— SBT-DF203-Lab7_202513549PF_OluyedeTobiSteven.pdf
- |— evidence/
 - |— [original PCAP]
 - |— fresh_dig_dns.pcapng
 - |— browser_dns.pcapng

- reports/
 - resolv_conf.txt
 - resolvectl_status.txt
 - dig_example_A.txt
 - dig_example_AAAA.txt
 - dig_example_MX.txt
 - dig_example_NS.txt
 - dns_capture_hashes.txt
 - fresh_dns_sha256.txt
 - dns_queries.tsv
 - dns_responses.tsv
 - browser_dns_inventory.txt
 - dns_A_answers.tsv
 - subsequent_tcp_destinations.tsv
- screenshots/
 - Screenshot_01_Evidence_Hash.png
 - Screenshot_02_Resolver_Configuration.png
 - Screenshot_03_Dig_A_Record.png
 - Screenshot_04_Dig_AAAA_Record.png
 - Screenshot_05_Dig_MX_Record.png
 - Screenshot_06_Dig_NS_Record.png
 - Screenshot_07_Fresh_DNS_Hash.png
 - Screenshot_08_DNS_Query_Fields.png
 - Screenshot_09_DNS_Response.png
 - Screenshot_10_Transaction_Correlation.png
 - Screenshot_11_Browser_DNS_Inventory.png
 - Screenshot_12_DNS_TCP_Correlation.png

```
(kali㉿kali)-[~/icdfaDF/GBT-DF203-Lab7]  
$ tree
```

```
├── evidence  
│   ├── browser_dns.pcapng  
│   ├── dig_dns.pcap  
│   └── fresh_dig_dns.pcapng
```

```
├── exported  
└── reports
```

```
    ├── browser_dns_inventory.txt  
    ├── dig_example_AAAA.txt  
    ├── dig_example_A.txt  
    ├── dig_example_MX.txt  
    ├── dig_example_NS.txt  
    ├── dig_example_short.txt  
    ├── dns_A_answers.tsv  
    ├── dns_capture_hashes.txt  
    ├── dns_queries.tsv  
    ├── dns_responses.tsv  
    ├── fresh_dns_sha256.txt  
    ├── resolv_conf.txt  
    ├── resolvectl_status.txt  
    ├── smtp_dns_correlation.tsv  
    └── subsequent_tcp_destinations.tsv
```

```
├── screenshots
```

```
├── scripts
```

```
└── working
```

```
    └── dig_dns_working.pcap
```

```
7 directories, 19 files
```

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Example Domains

described in RFC 2606 and RFC 6761, a number of domains such as
purposes. These domains may be used as illustrative examples in doc
registration or transfer.

We provide a web service on the example domain hosts to provide ba
provided as best effort, but are not designed to support production a
expected, please do not design applications that require the example

Further Reading

Managed Reserved Domains

Last revised 9 years ago.